

Contestant Number: _____

Time: _____

Rank: _____



DATABASE APPLICATIONS (240)

STATE – 2020

Production:

Printout 1: Property Table	_____	(60 points)
Printout 2: Enter Work Orders Printscreen	_____	(70 points)
Printout 3: Enter Work Orders Form (2 pages)	_____	(30 points)
Printout 4: Rent per Square Foot Query	_____	(40 points)
Printout 5: Rent per Square Foot Query + criteria	_____	(25 points)
Printout 6: City Property with Work Orders Report (2 pages)	_____	(70 points)
TOTAL POINTS	_____	(295 points)

Failure to adhere to any of the following rules will result in disqualification:

- 1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.**
- 2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.**
- 3. Electronic devices will be monitored according to ACT standards.**

No more than ten (10) minutes orientation
No more than ninety (90) minutes testing time
No more than ten (10) minutes wrap-up

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DATABASE APPLICATIONS SCORING SHEET

<i>Unless indicated otherwise, student receives all points or none.</i>	Points Possible	Points Earned
Printout 1 – Property Table (60 points maximum)		
Imported spreadsheet fields correctly (8 fields at 5 points a field)	40	
Sorted by City (ascending)	10	
Printed using best fit in landscape orientation (all data shows, 1 page)	10	
Misspellings or wrong data entered -1/2 pt per occurrence up to -10 max. Should also be 0 points for imported spreadsheet correctly.		
Printout 2 – Enter Work Orders PrintScreen (70 points maximum)		
Created a form with the correct fields (5 fields at 5 points a field).	25	
Property ID field is a combo box (signified by the down arrow on the printout)	10	
Repair Category is a list box (look at next printout field should be highlighted black)	10	
Word “completed” has been formatted to stand out.	5	
Two buttons (Add Record and Last Record) have been added. (5 points each)	10	
Form is laid out, so it is easy to read and understand.	10	
Printout 3 – Enter Work Orders Form (30 points maximum)		
Title is Enter Work Orders XX-XXXX-XXXX.	5	
Correct information was added to the form (4 records)	10	
Imported Work Orders spreadsheet and 3 records are visible in printout.	15	
Printout 4 – Rent per Square Foot Query (40 points maximum)		
Title is Rent per Square Foot Query XX-XXXX-XXXX	5	
Correct fields are included in the report (6 fields x 1 points each)	6	
Rent per Square Foot field calculations are correct. (this is all or nothing)	19	
Query sorted descending by rent per square foot field.	5	
Printed in landscape orientation, 1 page, all fields fully visible	5	
Printout 5 – Rent per Square Foot Query with CRITERIA (25 points max)		
Correct city is used as the criteria.	10	
All field titles and entries can be completely seen (no hidden or truncated)	10	
Printed in landscape orientation, 1 page.	5	
Printout 6 – City Property Report with Work Orders (70 points max)		
Title is City Property Report with Work Orders XX-XXXX-XXXX	5	
Correct fields are included in the report (7 fields x 2 points each)	14	
Report is grouped by City.	4	
Correct work order records are displayed on the report (16 records x 2 points each)	32	
Totals appear and are correct (3 totals at 5 points each) (Total for Property, Total for City, Total Monthly Rent)	15	
Total Points Earned	295	

General Instructions

You have been hired as a database specialist and will be assisting on various activities throughout Professional Business Associates, located at 5454 Cleveland Avenue, Columbus, Ohio 43231. Professional Business Associates provides accounting, software, programming and other services for clients.

1. You will have 90 minutes to complete your work.
2. Check that you have all the test pages.
3. Your name, initials, and school should *not* appear on any work you submit. Instead, type your contestant number everywhere XX-XXXX-XXXX appears.
4. If you finish before the end of the testing time, notify the administrator of the contest. Time may be a factor in determining a winner when there is a tie score.
5. When turning in your completed work, order the printed pages by job number, and place your scoring sheet on top of all your jobs. **Turn in all pages printed including incorrect printouts** (draw a line through each incorrect printout and place it at the back of the group).

Project: A customer needs assistance in developing a database to better manage their single-family rental homes. The customer is requesting a form to enter work orders as they are received and a customized report.

Create a database named **State-XX-XXXX-XXXX**, and save it as directed by the administrator.

Job 1: Property Table (Print 1)

1. Create a table by importing the spreadsheet named **SF Rental Properties** into Access. Accept the recommended field names. First row does contain headings. **Property ID** is indexed and no duplicates. Set Property ID as primary key. Name the table **Property Table XX-XXXX-XXXX**.
2. Make sure the fields have the following types:
 - a. Property ID = Number
 - b. Street Address = Short Text
 - c. City = Short Text
 - d. Zip = Number
 - e. Sq Footage = Number
 - f. Monthly Rent = Currency
 - g. Occupied = yes/no
3. Sort the table by City in ascending order.
4. Save the table.
5. Print the table in Landscape with all fields visible.

Job 2: Work Order Table and Form (one print)

1. Create a table named **Work Orders XX-XXXX-XXXX**. Use the same field names and assign the best data type for each field. **Work Order ID** is the primary key.
2. **Data Types:**
 - a. Work Order ID – autonumber (primary field)
 - i. NOTE: these will automatically number as you enter them.
 - b. Property ID – number (foreign key from the Properties table)
 - i. Field Size = Double
 - c. Repair Category – short text
 - d. Description – short text
 - e. Completed – yes/no
3. Create a **one-to-many relationship** between the Property and Work Order tables.
4. Use the **Form Wizard** to create a columnar form to enter work orders into this table.
 - a. Select all the fields from the Work Order table.
 - b. Name the form Enter Work Orders XX-XXXX-XXXX
5. In **Design View**, complete the following actions for Property ID:
 - a. Delete the Property ID label and text box.
 - b. Replace that with a combination box.
 - i. Choose the Property Table for values.
 - ii. Select the Property ID, Street Address and City fields.
 - iii. Sort ascending by City.
 - iv. Hide the key column and resize the columns so that all the information is visible.
 - v. Choose to store the value in the Property ID field.
 - vi. Name the combo box Property ID.
 - c. Resize and rearrange the label and Property ID combo box on the form.
 - d. Change the label to read “Property ID”.
6. In Design View, complete the following actions for Repair Category:
 - a. Delete the Repair Category label and text box.
 - b. Replace that with a list box.
 - i. Choose “I will type in the values I want”.
 - ii. Key in the following (do NOT key the number preceding the category):
 1. A/C
 2. Drywall
 3. Electrical
 4. Exterior
 5. Flooring
 6. Heat
 7. Other
 8. Plumbing
 9. Yard
 - iii. Choose to store the value in the Repair Category field.
 - iv. Name the combo box Repair Category.
 - c. Resize and rearrange the label and Repair Category list box on the form.
 - d. Change the label to read “Repair Category”.
7. Change the font size and color of “Completed” label so that it sticks out.

The screenshot shows a Microsoft Access form titled "Enter Work Orders xx-xxxx-xxxx". The form is in Design View. It contains the following controls:

- Work Order ID:** A text box with a "New" button next to it.
- Property ID:** A text box with a dropdown arrow.
- Repair Category:** A text box with a dropdown arrow, currently showing "A/C".
- Description:** A text box.
- Completed:** A label with a checkbox next to it.
- Buttons:** "Add Record" and "Last Record" buttons at the bottom.

8. Add at least two buttons (Add Record and Last Record) to the form, you may add more if you would like.
9. Should look like the example.
10. Open in Design View, use PrtScrn to capture the image, paste to a blank word document, resize so that the grader can read it and **PRINT**.

Job 3: Enter Work Order Form (one print that is 2 pages)

1. The customer provided the data shown below. Use the form to enter all records shown below (do NOT enter a work order ID it will be added automatically):

Work Order ID	Property Address	Repair Category	Description	Completed
	56 Windsor	Plumbing	Upstairs bathtub is not draining.	Yes
	118 Richardson	A/C	Fan is making a squealing noise.	
	46 Vale Road	Electrical	Lights flickering in the basement.	Yes
	292 Dogwood	Exterior	Paint is chipping on the front of the house.	

2. Import the Work Order spreadsheet to Append to the work order table.
 - a. NOTE: If you receive an Import Error table, delete it.
3. Print pages 1 and 2 of the Enter Work Order Form.

Job 4: Rent per Square Foot Query (one print)

1. Start a new query using the **Property Table XX-XXXX-XXXX** for the values.
2. Add the fields:
 - a. Property ID
 - b. Street Address
 - c. City
 - d. Sq Footage
 - e. Monthly Rent
3. Create a column to calculate the rent per square foot for each property.
 - a. Name the Column: Rent per Square Foot
 - b. HINT: Use builder to create an expression.
 - c. Sort in Descending Order
4. Print the query in landscape orientation.
5. NOTE: Notice which city has the **lowest** rent per square foot. You will need to know that city's name for the next job.

Job 5a: City Property Report with Work Orders (one print)

1. Open **Rent per Square Foot Query** in Design View.
 - a. From Job 4 you were supposed to determine which city had the lowest rent to square footage figures. Insert the city name in the Criteria field for the City field.
 - b. Run the query.
 - c. Print in landscape orientation with all the fields visible.

Job 5b: City Property Report with Work Orders (one print)

2. Start a new report using the report wizard. For fields choose all the fields from the **Rent per Square Foot query** and from the **Work Order Table XX-XXXX-XXXX** use only the Repair Category, Description and Completed fields.
 - a. View data with the Rent per Square Foot information on top.
 - b. Then choose "City" to group level.
 - c. Sort by Repair Category, ascending.
 - d. Choose a landscape orientation.
 - e. Name the report City Property Report with Work Orders XX-XXXX-XXXX.
3. View the report in **Layout View** to position the fields in an attractive manner and so all fields can be read.
 - a. Delete the Property ID label and field.
 - b. Rename Rent per Square Foot label to read "per SqFt".
 - c. Save the report.
4. View the report in **Design View** to calculate the total rental income for this city's properties.
 - a. HINT: Use the Totals button in the Design View to assist.
 - b. Add a label to the City footer to label the total rental income as:
 - i. Total Rent for the City
5. Still in **Design View**, have Access COUNT how many work orders have been issued for each property and in total for this city.
 - a. HINT: Use the Totals button in the Design View to assist.
 - b. At the Property Level add a label for the total work orders for the property as:
 - i. Total Work Orders for Property
 - c. Add a label to the City footer to label the total work orders as:
 - i. Total Work Orders for City
6. **Print** the report, be sure to use normal margins (it will be two pages).